Remove Duplicates From Linked List

AlgoExpert · Easy · Linked Lists

The problem

- List is sorted; remove nodes with repeated values in place.

Intuition

- The list is sorted, so duplicates are adjacent; we can skip them in place.

Brute force vs optimal

- With a hash set: works for unsorted lists, O(n) space
- Sorted list: duplicates are adjacent → O(1) space
- Rewire next pointers in place

Approach

- 1. Walk the list
- 2. While the next node has the same value, bypass it
- 3. Move to the next distinct node
- 4. Return the head

Worked example

original



result so far



Value 1: skip 1 duplicate(s)

Worked example

original



result so far



Value 3: skip 0 duplicate(s)

Worked example

original



result so far



Value 4: skip 2 duplicate(s)

Worked example

original



result so far



Value 5: skip 0 duplicate(s)

Worked example

original



result so far



Value 6: skip 1 duplicate(s)

Java solution

```java
public class RemoveDuplicatesFromLinkedList {
    public static LinkedListNode removeDuplicatesFromLinkedList(LinkedListNode head) {
        LinkedListNode cur = head;
        while (cur != null) {
            LinkedListNode next = cur.next;
            while (next != null && next.value == cur.value) next = next.next;
            cur.next = next;
            cur = next;
        }
        return head;
    }
}
```

Complexity

- Time: O(n)
- Space: O(1)

Edge cases & pitfalls

- Empty list or single node
- Runs of 3+ duplicates (loop, don't just skip one)
- Last node duplicates

Interview tips & follow-ups

- Follow-up: remove ALL nodes that had duplicates
- Follow-up: unsorted list

Recap

- The list is sorted, so duplicates are adjacent; we can skip them in place.
- Complexity: O(n)